

1. Network Layer Foundations & Switching

Mod 01

- **Role:** Host-to-Host end-to-end packet delivery across heterogeneous internetworks.
- **Store-and-Forward:** Router stores entire frame, verifies checksum, routes via table, and forwards.
- **Routing vs Forwarding:**
 - **Routing (Control Plane):** Software algorithms (OSPF, BGP) computing end-to-end optimal path.
 - **Forwarding (Data Plane):** Fast hardware engine transferring packet from input to output port.
- **Datagram vs Virtual Circuit (VC):**
 - *Datagram (Internet):* Connectionless, stateless routers, full 32-bit IP in every packet, dynamic reroute on failure, out-of-order delivery.
 - *Virtual Circuit (ATM/X.25):* 3-way handshake setup, stateful switches, 16-bit VCI tag, router crash kills circuit, guaranteed in-order.

3. IPv4 Fragmentation & MTU Mathematics

Mod 03

- **MTU (Maximum Transmission Unit):** Max L2 data payload (Ethernet MTU = 1500 Bytes).
- **Identification (16b):** Unique packet ID copied to all fragments for reassembly.
- **Flags (3b):**
 - Bit 0: Reserved (Must be 0).
 - Bit 1: **DF (Don't Fragment):** If 1 and packet > MTU, drop & send ICMP Type 3 Code 4.
 - Bit 2: **MF (More Fragments):** 1 = More fragments follow; 0 = Last fragment.

2. IPv4 Datagram Header Architecture (20-60B)

Mod 02

- **VER (4b):** Strictly 4 for IPv4 (0100 binary).
 - **HLEN (4b):** Header length in 4-byte words. Min = 5 (20B), Max = 15 (60B).
 - **ToS / DiffServ (8b):** Quality of Service priority bits.
 - **Total Length (16b):** Header + Payload in bytes. Max = 65,535B.
 - **TTL (8b):** Time-to-Live. Decrement by 1 at each router; at 0, packet dropped & ICMP Type 11 sent.
 - **Protocol (8b):** L4 multiplexing: 1=ICMP, 6=TCP, 17=UDP, 2=IGMP, 89=OSPF.
 - **Header Checksum (16b):** 1's complement sum of 16-bit words. Verifies header only; recomputed at every hop due to TTL.
- Payload Size = Total Length - (HLEN × 4) Bytes

4. IPv4 Classful Addressing Architecture & Special IP Blocks

Mod 04

Class	Leading Bits	1st Octet Range	NetID / HostID Bits	Default Mask	Networks / Hosts
A	0	1 – 126	8b / 24b	255.0.0.0 (/8)	126 / 16,777,214
B	10	128 – 191	16b / 16b	255.255.0.0 (/16)	16,384 / 65,534
C	110	192 – 223	24b / 8b	255.255.255.0 (/24)	2,097,152 / 254
D	1110	224 – 239	Multicast Group	N/A (No NetID/HostID)	Multicast addressing
E	1111	240 – 255	Reserved / Experimental	N/A	Research & Military

- **Private IPs (RFC 1918 - Non-routable):**
 - Class A: 10.0.0.0 – 10.255.255.255 (/8)
 - Class B: 172.16.0.0 – 172.31.255.255 (/12)
 - Class C: 192.168.0.0 – 192.168.255.255 (/16)
- **Special IP Addresses:**
 - 127.0.0.0/8 (127.0.0.1): Loopback testing.
 - 0.0.0.0: "This host on this network" (DHCP discover).
 - 255.255.255.255: Limited Broadcast (never routed).
 - NetID.All-1s: Directed Broadcast Address (DBA).

5. Subnetting Mechanics & Classful Design

Mod 05

- **Purpose:** Divides large wasteful host space into smaller departmental logical networks without changing external NetID.
- **Bit Borrowing:** Borrow s MSB bits from Host ID to form Subnet ID.
 - Subnets Formed = 2^s
 - Usable Hosts/Subnet = $2^h - 2$ (Subtract Subnet ID & DBA)
 - Block Size (Jump) = 256 - Interesting Mask Octet = 2^h
 - Router Subnet ID = Destination IP AND Subnet Mask
- **Example:** Class C 192.168.1.0 with 4 subnets $\implies s=2, h=6$. Mask = 255.255.255.192 (/26). Block size = 64. Subnets: .0, .64, .128, .192. Usable hosts = 62.

7. NAT & PAT (Port Address Translation)

Mod 07

- **NAT (RFC 1631):** Maps private RFC 1918 IPs to registered public IPs.
- **Types:**
 - *Static NAT:* 1:1 permanent mapping (Servers).
 - *Dynamic NAT:* Pool of public IPs assigned dynamically.
 - *PAT (NAT Overload):* Thousands of private IPs share **1 single public IP** using unique L4 Port numbers!
- **Translation Table:** Maps [Private IP:Port] $\rightarrow$ [Public IP:NAT Port].

9. Address Resolution Protocol (ARP) & RARP

Mod 08

- **Purpose:** Resolves Layer 3 Logical IP to Layer 2 Physical MAC.
- **ARP Request:** Broadcast (FF:FF:FF:FF:FF:FF): "Who has IP X? Tell Y".
- **ARP Reply:** Unicast back to requester: "I have IP X! My MAC is B".
- **ARP Cache:** RAM table storing (IP, MAC, TTL) to avoid repeated broadcasts. Dynamic entries age out (120-300s).

6. CIDR & Supernetting (Route Aggregation)

Mod 06

- **CIDR (RFC 1519):** Eliminates class boundaries. Notation IP/n.
- **3 Rules of CIDR Allocation:**
 1. Addresses in block must be contiguous.
 2. Total IPs $N = 2^{(32-n)}$ must be a power of 2.
 3. First address in decimal must be divisible by N .
- **Supernetting:** Aggregates contiguous Class C/CIDR routes into 1 entry (reduces global router tables). Prefix length decreases ($/24$ to $/22$).
- **Longest Prefix Match (LPM):** When multiple routes match a destination, router picks the one with the **longest prefix length** (most specific mask).

8. IPv6 Architecture & Transition

Mod 07

- **Address Space:** 128 bits (16 bytes) = 2^{128} approx 3.4 times 10^{38} IPs.
- **Base Header:** Fixed 40 Bytes. No checksum (removed), no broadcast (replaced by multicast/anycast).
- **8 Fields:** Version(4), Traffic Class(8), Flow Label(20), Payload Len(16), Next Hdr(8), Hop Limit(8), Src IP(128), Dst IP(128).
- **Transition Strategies:**
 - *Dual Stack:* Routers run both IPv4 and IPv6 stacks.
 - *Tunneling:* IPv6 encapsulated inside IPv4 packet.
 - *NAT-PT / Translation:* Header format translation gateways.

- **ARP 28B Packet:** HW Type (1=Eth), Prot Type (0x0800), HW Len (6), Prot Len (4), Opcode (1=Req, 2=Rep), Sender MAC/IP, Target MAC/IP.
- **Gratuitous ARP:** Host ARPs its own IP to detect IP conflicts or update neighbor caches on NIC change.
- **RARP:** MAC to IP resolution for diskless nodes (Replaced by DHCP).

10. Dynamic Host Configuration (DHCP)

Mod 09

- **Role:** Dynamic auto-assignment of IP, Mask, Gateway, DNS.
- **DORA Sequence (UDP Client:68, Server:67):**
 - **Discover:** Client broadcasts (255.255.255.255) from 0.0.0.0 .
 - **Offer:** Server proposes IP lease, mask, server ID.
 - **Request:** Client broadcasts accepting server's offer.
 - **ACK:** Server confirms lease with gateway and DNS info.
- **DHCP Relay Agent:** Router unicasts client broadcast across subnets.
- **Lease Timers:** T1 (50% time) unicast renew; T2 (87.5%) broadcast rebinding.

12. Distance Vector Routing & Count-to-Infinity

Mod 11

- **Algorithm:** Bellman-Ford: $D_x(y) = \min_v \{ c(x,v) + D_v(y) \}$.
- **RIP Protocol:** Periodic broadcast to immediate neighbors every 30s.
- **Count-to-Infinity Problem:** Link failure creates routing loops where nodes increment metric slowly to infinity (ping-pong bounce).
- **Mitigations:**
 - *Define Infinity* = 16 (Caps maximum loop count).
 - *Split Horizon:* Never advertise route back to neighbor learned from.
 - *Poison Reverse:* Advertise reverse route with metric = ∞ (16).

14. Path Vector (BGP) & Hierarchical Routing

Mod 13

- **BGP-4:** Inter-domain routing across Autonomous Systems (AS).
- **Path Vector:** Advertises complete AS-PATH: [AS200, AS100] .
- **Loop Discard Rule:** If router sees its own AS in AS-PATH, *discards update instantly!*
- **Hierarchical Routing:** Regions/clusters reduce tables from $O(N)$ to $O(\ln N)$.
Optimal Kamoun-Klein size: $S = 2 \cdot e \cdot \ln N$.

11. Internet Control Message Protocol (ICMP)

Mod 10

- **Role:** Network layer (Prot=1) error reporting and queries.
- **Error Types:**
 - *Type 3:* Destination Unreachable (Code 0=Net, 1=Host, 3=Port).
 - *Type 11:* Time Exceeded (Code 0=TTL Expired, 1=Reassembly).
 - *Type 12:* Parameter Problem. *Type 5:* Redirection.
- **Query Types:** Type 8 (Echo Req) & Type 0 (Echo Rep) $\implies$ ping .
- **Traceroute:** Sends packets with TTL=1, 2, 3... Intermediate routers drop packet and return ICMP Type 11. Target returns ICMP Type 3 Code 3.

13. Link State Routing & Dijkstra

Mod 12

- **Algorithm:** Dijkstra Shortest Path Tree (OSPF / IS-IS).
- **5 Phases:** (1) Discover neighbors via HELLO, (2) Measure link cost, (3) Build LSP, (4) Reliable Flooding of LSPs to all routers, (5) Run Dijkstra on complete LSDB.
- **DVR vs LSR:** DVR has local view and slow convergence (loops); LSR has global topology map, fast convergence, loop-free, but higher CPU/RAM overhead.

15. Congestion Control & Traffic Shaping

Mod 14

- **Open-Loop (Prevention):** Retransmission, window, drop policies.
- **Closed-Loop (Reaction):** Choke packets, backpressure, ECN bits.
- **Leaky Bucket:** Rigid constant output rate (Drops bursts).
- **Token Bucket:** Generates tokens at rate R \$, bucket capacity C \$. Allows bursts! Burst time: $T = C / (M - R)$.

16. Top 5 AKTU Semester Exam PYQs (10-Marks Long Questions)

Mod 14

Question Topic	Key Derivation / Diagram to Draw	Ref Module
Datagram vs Virtual Circuit	Setup phase, stateless vs stateful, VCI tag, router failure rerouting table	Module 01
IPv4 Header Format & Fields	Draw 32-bit grid: VER, HLEN, Total Len, Identification, Flags, Offset, TTL, Prot, Checksum	Module 02
Subnetting Numericals	Formula $2^s \geq N$, borrow bits, jump size $= 256 - \text{ext}\{\text{mask}\}$, first/last usable IP, broadcast	Module 05
Count-to-Infinity & Solutions	Linear 3-node graph, link failure loop increment, Split Horizon with Poison Reverse	Module 11
Leaky vs Token Bucket	Draw water bucket with hole vs token bucket with tokens; burst formula $T = C / (M - R)$	Module 14